

A Conditional Reduction for the L^q , $p = 1$ Endpoint

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Status

This is a conditional reduction packet for Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. The earlier packet

`1907.11588.Lq_general_local_characteristic_domination`

proves the $L^q(S)$ case for $1 < p, q < \infty$. This packet isolates the remaining endpoint $p = 1$, $1 < q < \infty$, to one explicit square-function comparison.

The endpoint comparison is not proved here. Thus the packet is conditional: if the endpoint square-function aggregate comparison in Lemma 1 is true for a fixed q , then the full $L^q(S)$ -valued characteristic-domination conclusion at $p = 1$ follows for that q .

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source states that the general local-martingale case is open and reduces the missing issue to an independent-increment comparison.

Here we use Proposition 12.8 and by taking $m = 1$ get from (12.11), (12.14), (2.9), and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n \geq 1}$ and $(\eta_n)_{n \geq 1}$ be X -valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exists a constant C (perhaps depending on p and X) such that

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C \mathbb{E} \left\| \sum_n \eta_n \right\|^p?$$

The continuation states the aggregate-domination formulation and notes that even the symmetric case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n 's and η_n 's are symmetric. But even the symmetric case is not known for the author.

Conditional Dependency

For a finite independent family $z = (z_i)$ of $L^q(S)$ -valued random variables, put

$$\mathrm{SF}_q(z) = \mathbb{E} \left\| \left(\sum_i |z_i|^2 \right)^{1/2} \right\|_{L^q(S)}.$$

Let

$$\mu_z = \sum_i \mathcal{L}(z_i)|_{L^q(S) \setminus \{0\}}$$

be the aggregate occurrence measure.

Endpoint Lemma 1 (Endpoint square-function aggregate comparison). *Let $1 < q < \infty$. There is a constant C_q such that for every measure space S and every two finite independent families $x = (x_i)$, $y = (y_j)$ of $L^q(S)$ -valued random variables,*

$$\mu_x \leq \mu_y \implies \mathrm{SF}_q(x) \leq C_q \mathrm{SF}_q(y).$$

This is the only unproved assertion used in the conditional theorem below. It is true in Hilbert spaces by the Laplace-transform argument in the companion Hilbert all- p packet, and the broad c_0 counterexample shows that no analog can hold in arbitrary Banach spaces. Finite-dimensional regrouping tests for fixed $q < \infty$ found no growing obstruction.

Idea of the Proof

At $p = 1$, the Dirksen–Yaroslavl'tsev Rosenthal norm still gives the upper bound for independent L^q -valued sums, but their printed two-sided theorem uses L^p -duality and is stated for $p > 1$. Yaroslavl'tsev's later BDG paper supplies the missing endpoint geometry in a different form: for UMD Banach function spaces, and hence for $L^q(S)$, the L^1 size of a martingale is equivalent to the L^1 size of its lattice square function.

Thus the $p = 1$ discrete bottleneck is exactly the question whether this lattice square function is stable under aggregate domination of the jump laws. If Lemma 1 holds, the discrete comparison follows at once from the $p = 1$ Banach-function-space BDG theorem. The passage from discrete independent increments to general local martingales is then the same total-characteristic reduction used in the existing $1 < p < \infty$ packet: continuous parts are already controlled at $p = 1$, and jump parts are compared only after accessible and quasi-left-continuous jumps are folded into the total spatial compensator.

Discrete Endpoint Consequence

Proposition 1 (Conditional discrete $p = 1$ comparison). *Assume Lemma 1 for a fixed $1 < q < \infty$. Let $(\xi_i)_{i=1}^n$ and $(\eta_j)_{j=1}^m$ be independent mean-zero $L^q(S)$ -valued random variables. If*

$$\sum_i \mathbb{P}(\xi_i \in B) \leq \sum_j \mathbb{P}(\eta_j \in B) \quad (B \in \mathcal{B}(L^q(S)), 0 \notin B),$$

then

$$\mathbb{E} \max_{1 \leq k \leq n} \left\| \sum_{i=1}^k \xi_i \right\|_{L^q(S)} \leq C_q \mathbb{E} \max_{1 \leq k \leq m} \left\| \sum_{j=1}^k \eta_j \right\|_{L^q(S)}.$$

In particular,

$$\mathbb{E} \left\| \sum_i \xi_i \right\|_{L^q(S)} \leq C_q \mathbb{E} \left\| \sum_j \eta_j \right\|_{L^q(S)}.$$

Proof. The aggregate hypothesis is exactly $\mu_\xi \leq \mu_\eta$, hence Lemma 1 gives

$$\mathbb{E} \left\| \left(\sum_i |\xi_i|^2 \right)^{1/2} \right\|_{L^q} \leq C_q \mathbb{E} \left\| \left(\sum_j |\eta_j|^2 \right)^{1/2} \right\|_{L^q}.$$

Yaroslavl'tsev's Banach-function-space BDG theorem [3, Theorem 4.32 and Proposition 4.33] says that for every $L^q(S)$ -valued martingale difference sequence $d = (d_i)$,

$$\mathbb{E} \sup_N \left\| \sum_{i=1}^N d_i \right\|_{L^q} \simeq_q \mathbb{E} \left\| \left(\sum_i |d_i|^2 \right)^{1/2} \right\|_{L^q}.$$

Both (ξ_i) and (η_j) , in any fixed order, are martingale difference sequences. Applying the BDG equivalence to the two families gives the maximal comparison. The terminal comparison follows by dropping the maximum. $\square$

Remark 1. *Dirksen and Yaroslavl'tsev's independent Rosenthal theorem [2, Theorem 2.1] explicitly keeps the upper bound at $p = 1$. Proposition 1 explains the missing lower side: at the endpoint the right object is not a raw $D_{q,q}$ or $D_{1,q}$ term, but the random lattice square function controlled by Yaroslavl'tsev's $p = 1$ BDG theorem.*

Conditional General-Local Consequence

Theorem 1 (Conditional L^q , $p = 1$ characteristic domination). *Fix $1 < q < \infty$ and assume Lemma 1. Let N and M be $L^q(S)$ -valued local martingales. If N is characteristically dominated by M in the sense of [1, Definition 12.9], then*

$$\mathbb{E} \sup_{t \geq 0} \|N_t\|_{L^q(S)} \leq C_q \mathbb{E} \sup_{t \geq 0} \|M_t\|_{L^q(S)}.$$

Proof. We first treat zero initial values. Since $L^q(S)$ is UMD, the source's canonical and decoupled tangent machinery applies. The continuous part is already controlled for all $0 < p < \infty$ by Yaroslavtsev's continuous characteristic-domination estimate [1, Proposition 12.6].

It remains to compare the purely discontinuous contribution. Conditional on the local characteristics, the decoupled tangent jump martingales have independent increments with deterministic total jump characteristics. After standard truncation of time and jump size, their jumps are finite independent families whose aggregate occurrence measures are the corresponding total spatial compensators. Characteristic domination gives

$$\nu^N(\mathbb{R}_+ \times B) \leq \nu^M(\mathbb{R}_+ \times B), \quad B \in \mathcal{B}(L^q(S)), \quad 0 \notin B.$$

Therefore Proposition 1, applied conditionally and then passed to the limit by monotone/localization approximation, gives the L^1 maximal comparison for the jump tangent martingales.

The existing $1 < p < \infty$ packet verified the important structural point needed here: the comparison uses only the total spatial jump compensator $\nu(\mathbb{R}_+ \times \cdot)$, not a separate domination of accessible and quasi-left-continuous parts. Time labels and the accessible/qlc tag can be erased for the identity jump integrand by conditional averaging over the spatial jump value and by Radon–Nikodym restriction of decompositions. The same erasure is built into Lemma 1, which is stated directly in terms of aggregate occurrence measures.

Combining the continuous and discontinuous estimates with the $p = 1$ canonical-decomposition equivalence from Yaroslavtsev's BDG paper [3, Section 4.3] yields the zero-start conclusion.

For nonzero initial values, Definition 12.9 gives $|\langle N_0, x^* \rangle| \leq |\langle M_0, x^* \rangle|$ for all x^* , hence $\|N_0\|_{L^q} \leq \|M_0\|_{L^q}$. Apply the zero-start result to $N - N_0$ and $M - M_0$, and use the triangle inequality. $\square$

Evidence and Failed Routes

Hilbert endpoint. For Hilbert spaces, Lemma 1 reduces to a scalar fractional-moment comparison for $(\sum_i \|\xi_i\|^2)^{1/2}$. The companion Hilbert all- p packet proves this by the Laplace identity

$$\mathbb{E}S^\alpha = c_\alpha \int_0^\infty (1 - \mathbb{E}e^{-tS})t^{-1-\alpha} dt, \quad 0 < \alpha < 1,$$

and the estimate

$$1 - \prod_i (1 - u_i(t)) \simeq \min\{1, \sum_i u_i(t)\}.$$

This argument uses only the aggregate law of the squared jump sizes.

c_0 boundary. The broad Banach-space statement is false in c_0 : deterministic coordinate jumps have supremum norm 1, while regrouping the same aggregate law by random coordinates creates an occupancy maximum of order $\log N / \log \log N$. This is exactly the obstruction one would expect to disappear at fixed finite q , where occupancy moments are finite constants depending on q .

Endpoint Rosenthal lower bound. A tempting route is to extend the Dirksen–Yaroslavtsev $s_{p,q}$ -norm equivalence to $p = 1$ by replacing $D_{p,q}$ with $D_{1,q}$. Their paper does state the independent upper estimate at $p = 1$, but the printed lower bound uses L^p -duality and does not reach the endpoint. Yaroslavtsev's $p = 1$ Banach-function-space BDG theorem supplies a square-function lower object, but I did not find a complete proof that the endpoint Rosenthal sum/intersection norm is always controlled by this square function.

Scalar disjointification literature. Johnson–Schechtman, Montgomery-Smith, and the later Jiao–Sukochev–Xie–Zanin disjointification theorems give sharp scalar formulas for

$$\left\| \left\| \sum_k f_k e_k \right\|_{\ell_q} \right\|_{L_p}$$

for all $0 < p, q < \infty$. These results strongly resemble the missing endpoint lemma. They do not immediately prove it because here each jump is an $L^q(S)$ -valued random function, so the dependence among spatial coordinates inside one jump is part of the problem.

Finite Checks

The script `code/grouping_square_simulation.py` randomly generates a finite aggregate law in ℓ_q^d , partitions the same atoms into two different independent families, and Monte Carlo estimates the two endpoint square functions. In several hundred random regroupings for

$$q = 1.1, 1.2, 1.5, 1.8, 2, 3, 5, 10, 20$$

the largest observed ratios were below 1.10. These checks are not proof; they only support the claim that the c_0 occupancy mechanism does not persist at fixed finite q .

Novelty and Literature Check

Cheap run indexes were searched for 1907.11588, `p=1, endpoint, Lq, square function, aggregate domination, and characteristic domination`. Existing nearby packets cover c_0 , Hilbert all finite p, ℓ_q at $p = q$, discrete L^q for $1 < p < \infty$, and full L^q general local martingales for $1 < p < \infty$. No local packet proves the $L^q, p = 1$ endpoint.

Literature inspected during this push included Yaroslavtsev [1], Dirksen–Yaroslavtsev [2], Yaroslavtsev’s BDG paper [3], Montgomery-Smith–Pruss [4], and the scalar disjointification theorem of Jiao–Sukochev–Xie–Zanin [5]. I found strong nearby tools but no statement that directly proves Lemma 1.

Human Review Recommendation

Review the packet as a reduction, not as a solved endpoint theorem. The highest-value next task is to prove or disprove Lemma 1. Promising proof routes are:

- a vector-valued Johnson–Schechtman disjointification theorem for the mixed norm $L^1(\Omega; L^q(S; \ell^2))$;
- a Bernoulli–Poisson comparison for the L^q -lattice square function;
- an endpoint Davis decomposition proving the $p = 1$ Rosenthal sum/intersection norm is controlled by Yaroslavtsev’s square function.

References

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